

Exercise 1: Probability Density Functions and Monte Carlo generators

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Problem 1

Draw the Gaussian distribution for three different sets of values for the mean and standard deviation parameters (μ, σ) and explain how μ and σ parameters affect its shape.

If a newly discovered particle X is described by a Gaussian distribution $Gauss(\mu = 200\text{GeV}, \sigma = 2\text{GeV})$, calculate the probability to produce particle X with mass of 205 GeV or more. What is the probability to independently produce two particles X with masses above 203 GeV?

Problem 2

Let's make sure that the Central Limit Theorem works even with the exponential PDF $E(x; \lambda) = \lambda e^{-\lambda x}$. Generate a random number x which is defined as $x = \frac{1}{3}(x_1 + x_2 + x_3)$, where x_i is a random number drawn from $E(x; \lambda_i)$. Check the distribution of x after generating a large number of random events. Does it look like a Gaussian? Now redo the Monte Carlo experiment by defining x as $x = \frac{1}{N} \sum_{i=1}^N x_i$ and find N for which the resulting distribution looks like a Gaussian.

Problem 3

Let's study a fair economy design using the Monte Carlo methods. Design a society with 1000 inhabitants, each one starting with 100 coins. Each day, everyone randomly chooses a partner and they flip a coin. The loser has to pay 1 coin to the winner, unless they have no coins (in that case nothing happens). Study how the distribution of coins changes after 1, 5, 10, and 20 years. Plot the evolution of the mean and standard deviation of the coin distribution as a function of time. What do these numbers represent? Did you expect these results?

Problem 4

It surely can't be fair to have poor and rich people just by chance! Think of at least 3 different sets of rules (laws ;)) that you can introduce in your society from Problem 3 that could help fix this problem and result in a fair distribution of coins as time progresses.

Problem 5*

Build a society and study how it evolves by simulating birth, death, and limited resources. Start simple, defining a probability for each creature to give birth and to die and study how the population grows. Continue by defining a maximum capacity of the population (to simulate limited resources) and adjust the birth rate to respect it. Include the modeling of death probability dependence on the age. Can you replicate the age distribution of humans? What else can you introduce to make your simulation more realistic?